SUMMER INTERNSHIP PROJECT REPORT

Time-Series Analysis of COVID-19 Data

Submitted By

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Contents

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Abstract

The COVID-19 pandemic has generated an unprecedented volume of data, providing researchers with valuable opportunities to apply statistical and machine learning techniques for understanding disease dynamics and extracting meaningful insights. The availability of daily records of confirmed cases, deaths, and related indicators has enabled comprehensive analyses of temporal patterns, regional variations, and forecasting of future trends. This project aims to demonstrate the application of data science, statistical analysis, time-series forecasting, and unsupervised machine learning techniques using real-world datasets.

The primary dataset utilized in this study is the WHO COVID-19 Global Daily Dataset, which contains daily records of COVID-19 cases reported by countries and regions across the world. The data were subjected to preprocessing, cleaning, aggregation, and exploratory analysis to investigate the evolution of the pandemic. Various visualizations were employed to examine trends in global case counts, regional distributions, and temporal fluctuations. Time-series analysis techniques, including moving average analysis, growth rate analysis, seasonal decomposition, and stationarity assessment through the Augmented Dickey–Fuller test, were used to understand the underlying characteristics of the series.

To forecast future COVID-19 cases, three forecasting approaches were implemented: ARIMA, SARIMA, and Holt–Winters Exponential Smoothing. Model parameters were selected using ACF and PACF analysis, and model performance was evaluated using information criteria such as AIC and BIC. Comparative analysis showed that the Holt–Winters model provided the best overall fit among the forecasting models considered, highlighting the importance of incorporating seasonal effects in epidemiological forecasting.

In addition to the COVID-19 analysis, the project includes a machine learning component using the MNIST handwritten digit dataset. Unsupervised learning techniques, namely K-Means Clustering and Hierarchical Clustering, were applied to identify natural groupings within the data. Principal Component Analysis (PCA) was employed for dimensionality reduction and visualization of cluster structures. The quality of clustering was assessed using the macro-averaged F1 score, providing a quantitative measure of clustering performance.

Overall, the project demonstrates how statistical analysis, forecasting methodologies, dimensionality reduction, and clustering techniques can be combined to extract meaningful information from complex datasets. The findings emphasize the importance of data-driven approaches in understanding real-world phenomena and illustrate the practical utility of modern analytical tools in public health and pattern recognition applications.

Introduction

The rapid growth of data collection technologies has led to the generation of large and complex datasets across numerous domains, including healthcare, epidemiology, finance, and education. Extracting useful information from such data requires the application of statistical methods, machine learning techniques, and computational tools capable of identifying patterns, trends, and relationships. The field of data science integrates these approaches to transform raw data into actionable knowledge and support evidence-based decision-making.

The COVID-19 pandemic represents one of the most significant global health crises in recent history. Monitoring and analysing COVID-19 data is essential for understanding the progression of the disease, evaluating intervention strategies, and supporting public health planning. Time-series analysis provides a powerful framework for studying the temporal evolution of pandemic-related indicators and forecasting future developments based on historical observations.

In this project, the WHO COVID-19 Global Daily Dataset is analysed to investigate the temporal behaviour of COVID-19 cases worldwide. The study involves data preprocessing, exploratory data analysis, trend analysis, seasonal decomposition, and forecasting using classical time-series models. Particular emphasis is placed on comparing different forecasting approaches, including ARIMA, SARIMA, and Holt–Winters Exponential Smoothing, to determine the most suitable model for the given data.

Apart from forecasting, the project also explores unsupervised machine learning techniques through a clustering study on the MNIST handwritten digit dataset. Clustering methods are valuable tools for discovering hidden structures within data when class labels are unavailable. K-Means Clustering and Hierarchical Clustering are implemented to group similar observations, while Principal Component Analysis (PCA) is used to facilitate visualization and interpretation of the resulting clusters.

By integrating statistical analysis, forecasting methods, dimensionality reduction, and clustering techniques, this project provides a comprehensive demonstration of modern data-analytic methodologies. The study illustrates how different analytical approaches can be applied to diverse datasets in order to uncover meaningful insights, generate forecasts, and identify underlying structures that may not be immediately apparent from the raw data.

Objectives

The primary objective of this project is to analyse global COVID-19 data using statistical and machine learning techniques and to derive meaningful insights from the observed patterns and trends. The specific objectives of the study are as follows:

- To import, clean, and preprocess the WHO COVID-19 Global Daily Dataset for further analysis.
- To study the temporal evolution of COVID-19 cases and deaths across different countries and WHO regions.
- To identify the most affected countries based on cumulative COVID-19 cases.
- To aggregate and analyse COVID-19 data on daily, monthly, and quarterly bases.
- To compare the burden of COVID-19 among different WHO regions using descriptive statistical methods.
- To visualise trends and patterns in COVID-19 cases and deaths through appropriate graphical representations.
- To apply time-series analysis techniques for understanding the behaviour of COVID-19 data over time.
- To compute moving averages and growth rates to identify trends and fluctuations in the pandemic.
- To implement the K-Means clustering algorithm on the MNIST handwritten digit dataset.
- To evaluate clustering performance using the macro-averaged F1 score.
- To visualise and interpret clustering results using dimensionality reduction techniques such as Principal Component Analysis (PCA).
- To demonstrate the practical applications of statistics, data science, and machine learning in solving real-world problems.

Methodology

Overview

The methodology adopted in this project consists of a systematic sequence of steps involving data collection, data preprocessing, exploratory data analysis, time-series analysis, and clustering analysis. The overall workflow was implemented using Python and various data science libraries such as Pandas, NumPy, Matplotlib, Scikit-Learn, and Statsmodels.

Data Collection

The primary dataset used in this study was the WHO COVID-19 Global Daily Dataset. The dataset contains daily records of COVID-19 cases and deaths reported by countries and territories worldwide. In addition, the MNIST handwritten digit dataset available in the Scikit-Learn library was used for the clustering component of the project.

Data Preprocessing

Data preprocessing was performed to ensure consistency and accuracy before conducting the analysis. The following steps were carried out:

- Importing the COVID-19 dataset into Python using the Pandas library.
- Selecting relevant variables required for the analysis.
- Converting the `Date_reported` variable into datetime format.
- Filtering observations within the specified study period.
- Creating additional variables such as quarters and month-year combinations for aggregation purposes.
- Checking the dataset for inconsistencies and ensuring proper formatting of variables.

Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to understand the structure and characteristics of the dataset. Various descriptive statistics and graphical techniques were employed to identify trends and patterns.

The following analyses were performed:

- Daily global COVID-19 case analysis.
- Daily global COVID-19 death analysis.
- Identification of the most affected countries based on cumulative cases.
- WHO region-wise comparison of total cases.
- Quarterly aggregation of cases and deaths.
- Monthly comparison of cases across WHO regions.

Several graphical representations, including line plots, bar charts, pie charts, and heatmaps, were used to visualize the findings.

Time-Series Analysis and Forecasting

Since COVID-19 data are recorded sequentially over time, time-series techniques were employed to analyse temporal patterns and forecast future case counts.

The following procedures were performed:

- Aggregation of daily global COVID-19 case counts.
- Computation of 7-day moving averages to smooth short-term fluctuations.
- Analysis of daily growth rates to study changes in infection levels.
- Seasonal decomposition to separate trend, seasonal, and irregular components.
- Stationarity assessment using the Augmented Dickey–Fuller (ADF) test.
- Examination of autocorrelation and partial autocorrelation through ACF and PACF plots.
- Forecasting using ARIMA, SARIMA, and Holt–Winters Exponential Smoothing models.
- Comparison of forecasting models using AIC and BIC criteria.

These techniques provided insights into the temporal dynamics of the pandemic and enabled the identification of an appropriate forecasting model.

Clustering Analysis

To demonstrate unsupervised machine learning techniques, clustering analysis was performed using the MNIST handwritten digit dataset.

The following clustering methods were implemented:

- K-Means Clustering.
- Hierarchical Agglomerative Clustering.

The clustering procedure involved the following steps:

- Loading and preprocessing the MNIST dataset.
- Standardization of the feature matrix.
- Application of K-Means Clustering.
- Application of Hierarchical Clustering using Ward linkage.
- Assignment of observations to clusters.
- Mapping cluster labels to true digit labels.
- Evaluation of clustering performance using the macro-averaged F1 score.

Dimensionality Reduction and Visualization

Principal Component Analysis (PCA) was applied to reduce the dimensionality of the MNIST dataset and facilitate visualization of clustering results. The transformed two-dimensional representation was used to visualize cluster separation and interpret the structure of the high-dimensional data.

In addition, dendrograms were constructed for Hierarchical Clustering to illustrate the hierarchical relationships among observations and provide insight into cluster formation.

Software and Tools Used

The entire analysis was conducted using Python in the Google Colab environment. The major libraries used are listed below:

Library	**Purpose**
Pandas	Data manipulation and preprocessing
NumPy	Numerical computations
Matplotlib	Data visualization
Seaborn	Statistical graphics and heatmaps
Scikit-Learn	Clustering and machine learning
Statsmodels	Time-series modelling and forecasting

Table 1: Software libraries used in the project

Summary

The methodology combined statistical analysis, time-series modelling, and machine learning techniques to investigate COVID-19 data and demonstrate clustering applications. The systematic approach adopted in this study enabled the extraction of meaningful insights from large and complex datasets.

Data Analysis and Results

Exploratory Data Analysis

Daily Global COVID-19 Cases

The total number of new COVID-19 cases reported globally for each day was calculated by aggregating the daily case counts across all countries. This analysis helps in understanding the progression of the pandemic over time and identifying major waves of infection.

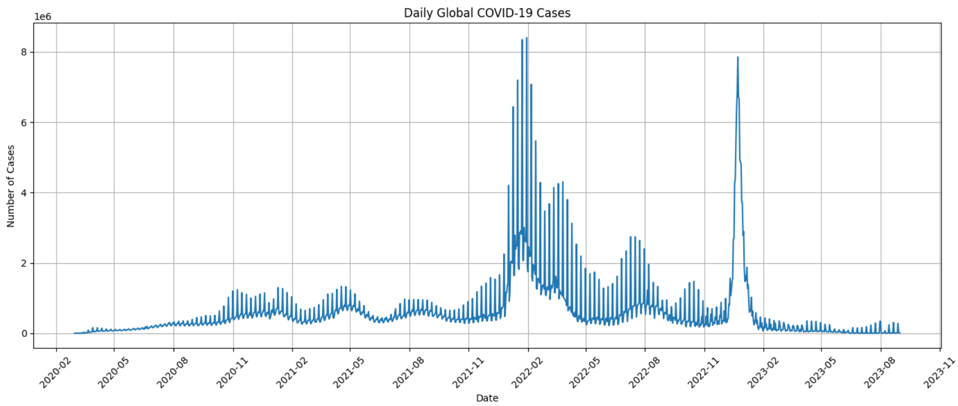

Figure 1: Daily Global COVID-19 Cases

The plot shows significant fluctuations in the number of daily cases, with multiple peaks corresponding to different waves of the pandemic.

Daily Global COVID-19 Deaths

Daily global deaths were obtained by summing the reported deaths across all countries for each day.

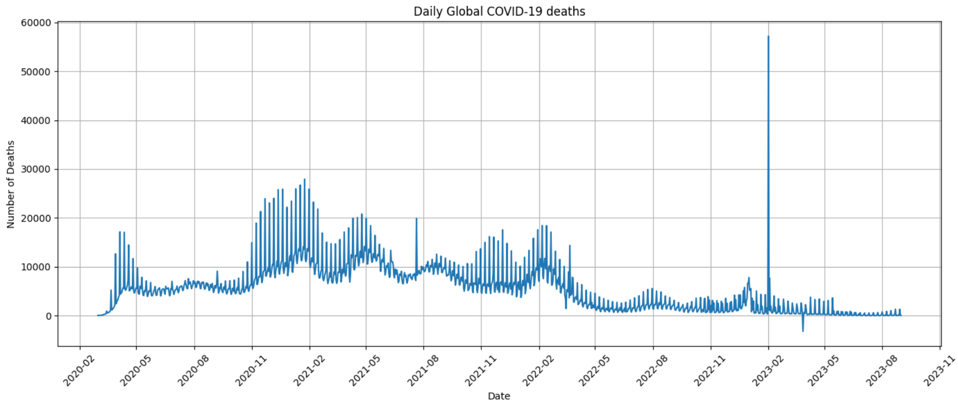

Figure 2: Daily Global COVID-19 Deaths

The trend in daily deaths generally follows the trend in daily cases, although with a time lag due to the progression of the disease.

Top Five Most Affected Countries

The countries with the highest cumulative number of COVID-19 cases were identified using the maximum reported cumulative cases.

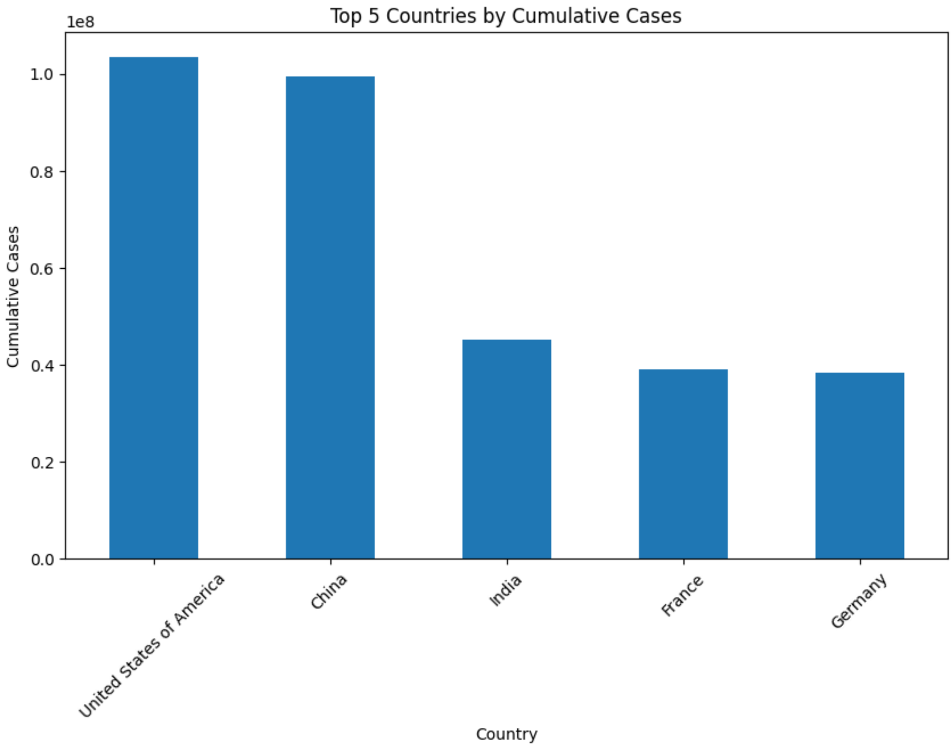

Figure 3: Top Five Countries by Cumulative Cases

The analysis reveals the countries that experienced the highest burden of COVID-19 infections during the study period.

WHO Region-wise Analysis

To compare the impact of COVID-19 across geographical regions, the total number of new cases was calculated for each WHO region.

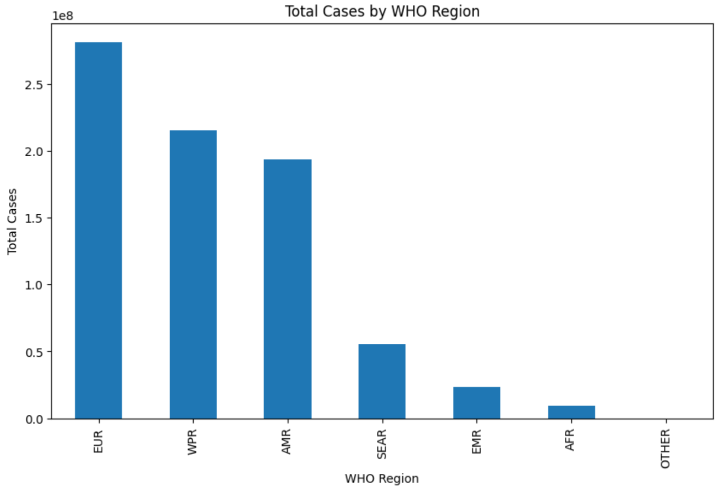

Figure 4: Total COVID-19 Cases by WHO Region

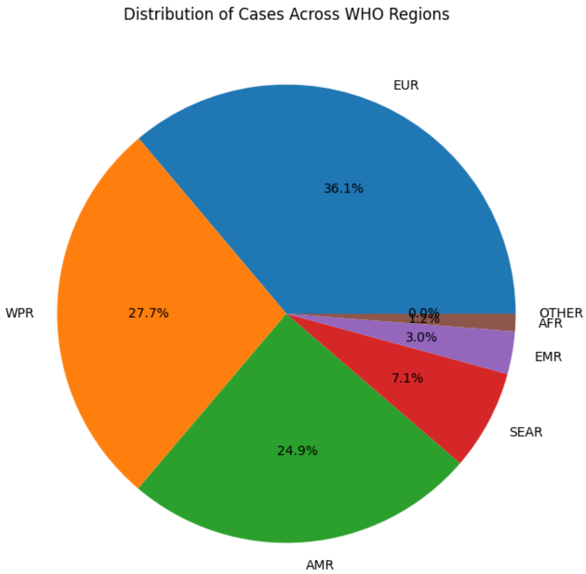

Figure 5: Pie Chart of total COVID-19 Cases by WHO Region

The comparison highlights regional differences in disease spread and reporting patterns.

Quarterly Analysis of Cases and Deaths

The dataset was aggregated on a quarterly basis to study the evolution of the pandemic over longer time intervals.

Table 2: Quarterly COVID-19 Cases and Deaths

Quarter	Total Cases	Total Deaths
2020 Q1	697567.0	40666.0
2020 Q2	9377665.0	511574.0
2020 Q3	23887893.0	546584.0
2020 Q4	48287242.0	834992.0
2021 Q1	45249877.0	1026874.0

Quarterly aggregation helps in identifying major phases of the pandemic and understanding long-term trends.

Monthly Cases by WHO Region

A pivot table was created to summarize monthly COVID-19 cases for each WHO region.

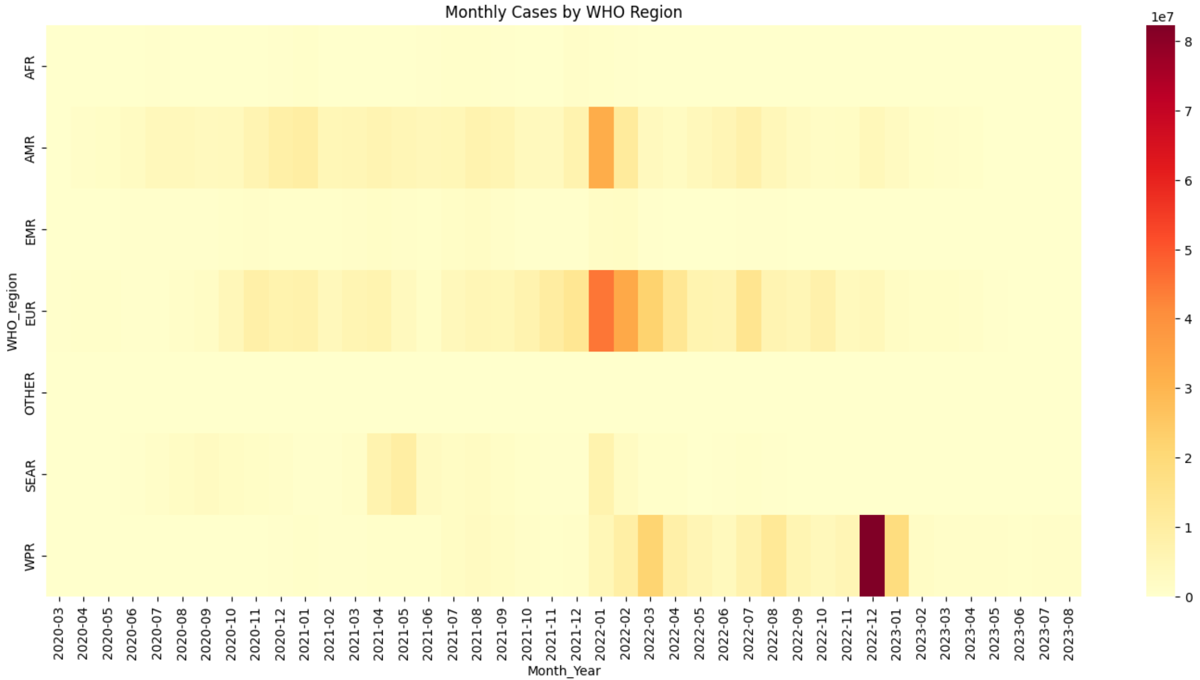

Figure 6: Monthly COVID-19 Cases by WHO Region

This analysis provides insights into how the pandemic evolved differently across regions over time.

Time Series Analysis

Seven-Day Moving Average

Daily COVID-19 data often exhibit significant short-term fluctuations due to reporting delays and random variation. To obtain a smoother representation of the underlying trend, a seven-day moving average was calculated.

The moving average is defined as:

$$MA_t = \frac{1}{7}\sum_{i=0}^{6} X_{t-i}$$

where X_t denotes the number of cases reported on day t.

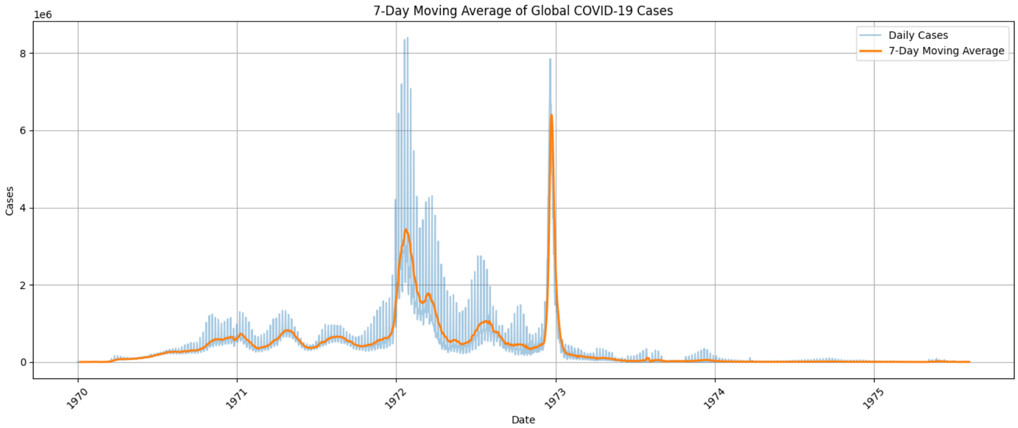

Figure 7: Seven-Day Moving Average of Global COVID-19 Cases

The moving average smooths short-term variability and facilitates identification of long-term trends and pandemic waves.

Growth Rate Analysis

The daily growth rate of COVID-19 cases was calculated to measure the relative change in case counts from one day to the next.

The growth rate is computed as:

$$Growth\ Rate = \frac{X_t - X_{t-1}}{X_{t-1}} \times 100$$

where X_t represents the number of cases on day t.

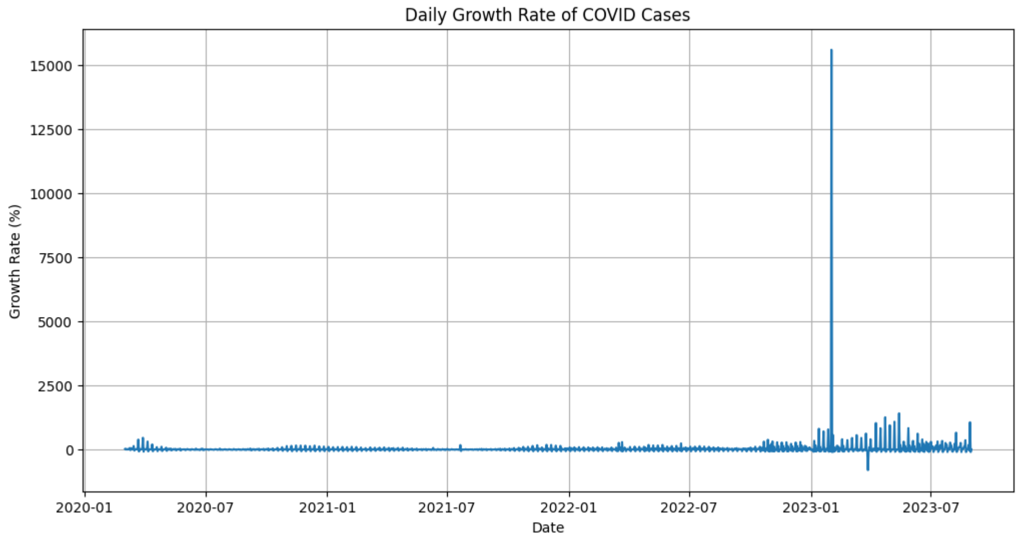

Figure 8: Daily Growth Rate of COVID-19 Cases

The growth rate analysis helps identify periods of rapid increase or decline in infection levels.

Seasonal Decomposition

Time-series data can often be decomposed into three major components:

- Trend Component
- Seasonal Component
- Residual Component

Seasonal decomposition was performed to separate these components and better understand the structure of the COVID-19 time series.

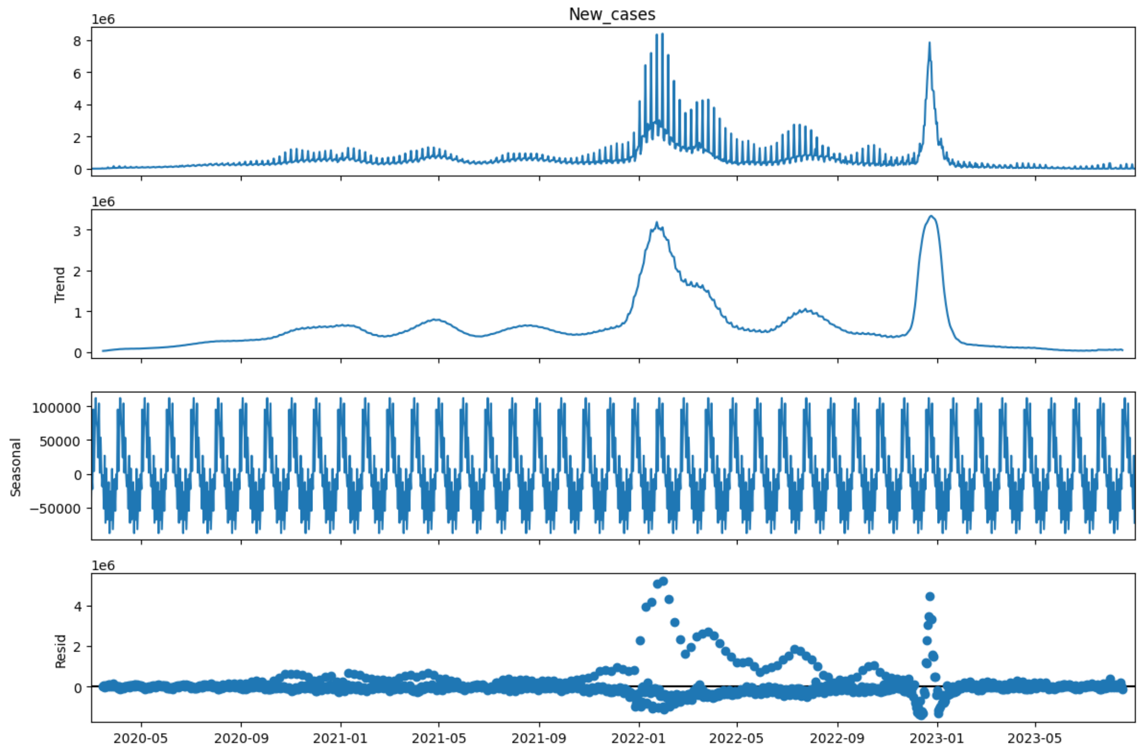

Figure 9: Seasonal Decomposition of Daily Global Cases

The decomposition provides insights into long-term trends and recurring patterns present in the data.

Results

- The daily global COVID-19 case series exhibited multiple waves of infection, with major peaks observed during 2022 and early 2023, indicating periods of intense disease transmission.
- The 7-day moving average successfully smoothed short-term fluctuations and clearly highlighted the underlying long-term trend of the pandemic.
- Growth rate analysis revealed substantial variability in daily case counts, with sharp increases during outbreak periods and declines during recovery phases.
- Seasonal decomposition identified distinct trend, seasonal, and residual components, demonstrating that the COVID-19 time series contains both systematic patterns and random fluctuations.

Stationarity Testing

Augmented Dickey–Fuller Test (ADF Test)

The Augmented Dickey–Fuller (ADF) test is used to determine whether a time series is stationary or contains a unit root. Stationarity is an important assumption for many time-series models.

- **Null Hypothesis (H_0):** The time series contains a unit root (non-stationary).
- **Alternative Hypothesis (H_1):** The time series does not contain a unit root (stationary).
- **Mathematical Model:**

$$\Delta Y_t = \alpha + \beta t + \gamma Y_{t-1} + \sum_{i=1}^{p} \delta_i \Delta Y_{t-i} + \varepsilon_t$$

  where,

  - Y_t: Value of the time series at time t.
  - ΔY_t: First difference of the series.
  - α: Constant term.
  - βt: Time trend.
  - γ: Coefficient of the lagged variable.
  - ε_t: Random error term.

- **Test Statistic:**

$$ADF = \frac{\hat{\gamma}}{SE(\hat{\gamma})}$$

  where $\hat{\gamma}$ is the estimated coefficient and $SE(\hat{\gamma})$ is its standard error.

- **Decision Rule:**

  Reject H_0 if the p-value is less than the chosen significance level (usually 0.05). In that case, the series is considered stationary.

Table 3: ADF Test Results

Statistic	**Value**
ADF Statistic	-3.8557
p-value	0.0024
1% Critical Value	-3.4356
5% Critical Value	-2.8638
10% Critical Value	-2.5680
Decision	Reject H_0
Conclusion	Series is Stationary

Interpretation:

The Augmented Dickey–Fuller (ADF) test produced a test statistic of -3.8557 with a p-value of 0.0024. Since the p-value is less than the 5% significance level, the null hypothesis of non-stationarity is rejected. Furthermore, the ADF statistic is smaller than the critical values at the 1%, 5%, and 10% significance levels. Therefore, the series may be regarded as stationary and no differencing is required before model fitting.

Autocorrelation Function (ACF)

The Autocorrelation Function (ACF) was used to identify the order of the moving average component of the ARIMA model. The ACF plot gives a comprehensive picture of how current values connect to past ones across a number of time steps by displaying the correlation between the time series and its lagged values.

- **Purpose:** The ACF plot helps identify the degree of correlation between observations at different lags. It is useful for determining the MA (Moving Average) terms in ARIMA models.If the ACF plot shows significant correlation at lag k and then cuts off, this suggests that q = k.
- **Interpretation:** The x-axis shows the number of lags. The y-axis shows the autocorrelation values (ranging from-1 to 1). Significant spikes outside the confidence interval (dashed lines) suggest a strong correlation at that particular lag. For example, if a spike at lag 1 is significant but the correlation quickly drops to near zero for later lags, it suggests that only the immediate previous value (lag 1) is important for predicting the current value.

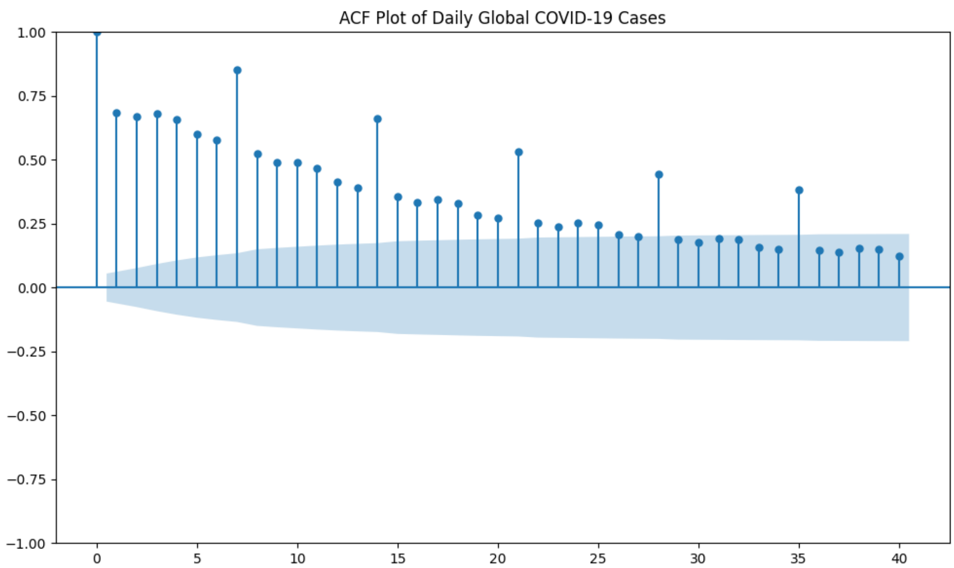

Figure 10: Autocorrelation Function (ACF)

Inference from ACF Plot:

The Autocorrelation Function (ACF) plot shows strong positive autocorrelations at several lags, indicating a significant dependence between current and past observations. The autocorrelation coefficients decline gradually rather than exhibiting a sharp cut-off, suggesting the presence of an autoregressive component in the series. Significant spikes are also observed at lags 7, 14, 21, 28, and 35, indicating a possible weekly seasonal pattern in the COVID-19 case data. The slow decay of the ACF suggests that a simple moving average model may not adequately capture the underlying structure of the series.

Partial Autocorrelation Function (PACF)

The Partial Autocorrelation Function (PACF) was used to identify the order of the autoregressive component.The PACF plot eliminates the impact of lesser lags and displays the correlation between a time series and its lagged values.

- **Purpose:** PACF helps in identifying the AR (AutoRegressive) terms in ARIMA models.If the PACF plot shows significant correlation at lag k and then cuts off, this suggests that p = k. It explains the direct relationship between a time series and its lag, removing the influence of any intermediate lags.
- **Interpretation:** Like the ACF plot, the x-axis represents the lags and the y-axis represents the partial autocorrelations. A spike at a given lag indicates a significant direct relationship between the time series and its lagged version, after accounting for the influence of previous lags. The first significant spike in the PACF plot can help determine the number of AR terms.

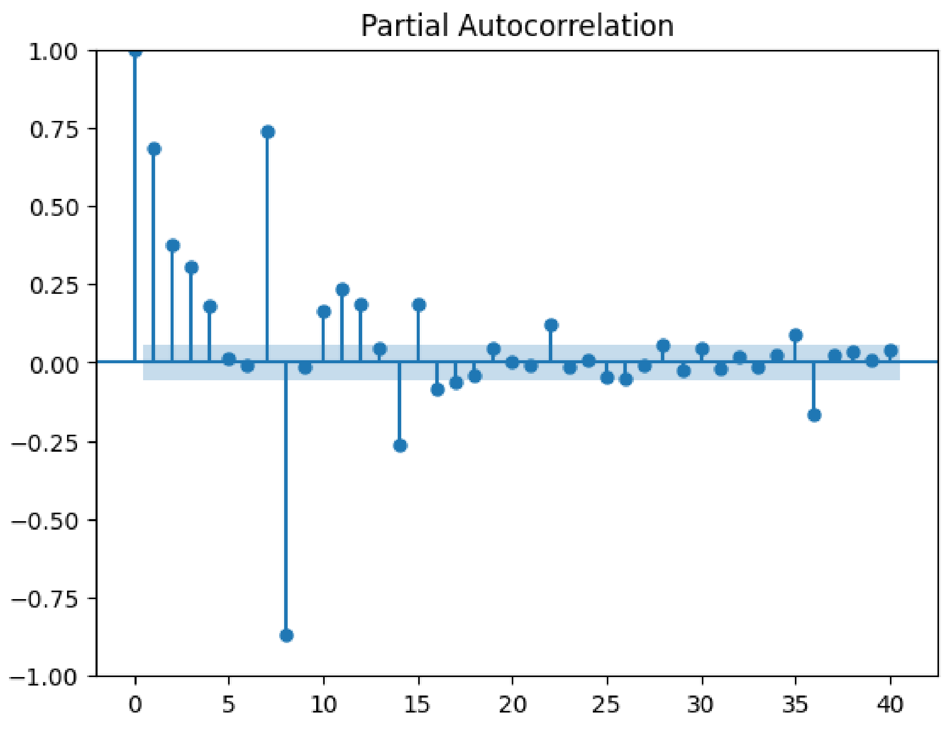

Figure 11: Partial Autocorrelation Function (PACF)

Inference from PACF Plot:

The Partial Autocorrelation Function (PACF) plot exhibits significant spikes at the first few lags, particularly at lags 1, 2, 3, and 7. After these initial lags, most of the partial autocorrelations lie within the confidence bounds, indicating a substantial reduction in direct dependence. The prominent spike at lag 7 further supports the existence of weekly seasonality in the data. The PACF behaviour suggests that a low-order autoregressive model may be appropriate for capturing the dependence structure of the series.

AutoRegressive Integrated Moving Average (ARIMA)

ARIMA (AutoRegressive Integrated Moving Average) is one of the most widely used models for time-series forecasting. It combines autoregressive (AR), differencing (I), and moving average (MA) components to model temporal dependence in a stationary time series.

Notation:

$$ARIMA(p, d, q)$$

where,

- p: Order of the autoregressive component.
- d: Degree of differencing.
- q: Order of the moving average component.

Mathematical Representation:

$$\phi(B)(1 - B)^d Y_t = \theta(B)\varepsilon_t$$

where,

- Y_t: Observed time series.
- B: Backshift operator.
- $\phi(B)$: Autoregressive polynomial.
- $\theta(B)$: Moving average polynomial.
- ε_t: White noise error term.

The ARIMA model is suitable for modelling non-seasonal time-series data and is commonly used for forecasting future observations. ‘‘‘

The Augmented Dickey–Fuller (ADF) test yielded a p-value of 0.0024, which is less than the significance level of 0.05. Therefore, the null hypothesis of non-stationarity was rejected, indicating that the series is stationary. Consequently, no differencing was required and the differencing order was chosen as $d = 0$.

The PACF plot exhibited significant spikes up to approximately lag 3, after which most of the values fell within the confidence bounds. This suggests the presence of an autoregressive component of order 3, leading to the selection of $p = 3$.

The ACF plot showed a gradual decay rather than a sharp cut-off, indicating the presence of a moving average component. A first-order moving average term was considered sufficient, resulting in $q = 1$.

Hence, based on the stationarity test and the behaviour of the ACF and PACF plots, the following model was selected:

$$ARIMA(3, 0, 1)$$

The adequacy of the selected model was further evaluated using model diagnostics and information criteria such as AIC and BIC.

Model Summary

Table 4: ARIMA(3,0,1) Model Statistics

Statistic	**Value**
Model	ARIMA(3,0,1)
Number of Observations	1279
Log-Likelihood	-18747.509
AIC	37507.017
BIC	37537.940
HQIC	37518.629

Forecast Plot

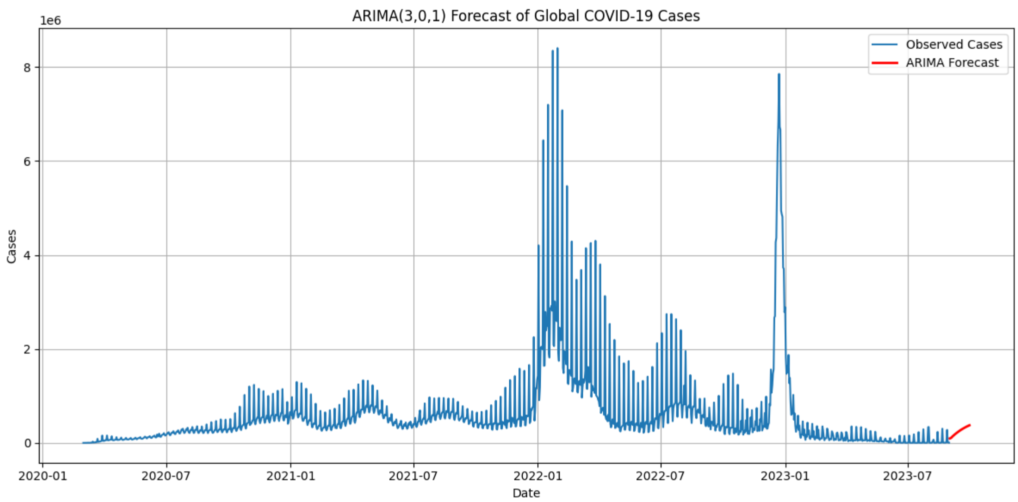

Figure 12: ARIMA Forecast

Interpretation:

Figure X presents the 30-day forecast obtained from the ARIMA(3,0,1) model fitted to the daily global COVID-19 case series. The blue line represents the observed number of daily cases, while the red line represents the forecasted values.

The historical series exhibits multiple waves of infection, with major peaks occurring during 2022 and early 2023. These peaks reflect periods of rapid disease transmission followed by gradual declines. The ARIMA model successfully captures the overall temporal dependence present in the data.

The forecast indicates a modest upward trend in the number of daily COVID-19 cases over the forecasting horizon. No abrupt increases or decreases are predicted, suggesting relatively stable future behaviour based on the historical pattern observed in the data. The forecast values remain substantially lower than the major pandemic peaks observed during the study period.

Overall, the ARIMA(3,0,1) model provides a reasonable short-term forecast of global COVID-19 cases and demonstrates the usefulness of time-series modelling techniques for understanding and predicting epidemiological trends.

Residual Analysis

Residual analysis was conducted to evaluate the adequacy of the fitted ARIMA model.

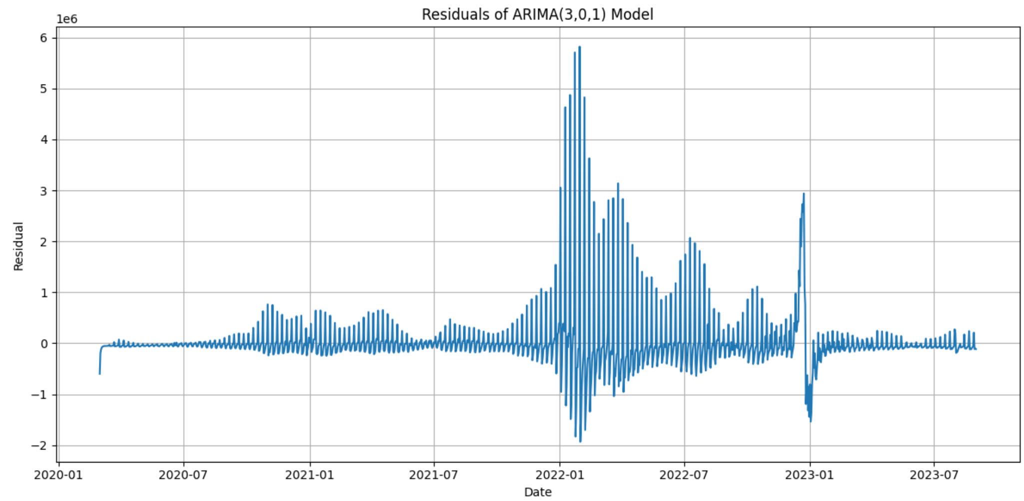

Figure 13: Residual Plot of ARIMA Model

Interpretation:

The residual series fluctuates around zero for most of the study period, indicating that the ARIMA model captures a substantial portion of the temporal structure present in the data. However, several large residuals are observed during major pandemic waves, particularly in early 2022 and early 2023. These extreme values correspond to sudden surges in reported COVID-19 cases and suggest that some irregular fluctuations remain unexplained by the model.

The Ljung–Box test was used to assess residual autocorrelation.

- Null Hypothesis (H_0): The residuals are independently distributed (no autocorrelation exists in the residuals).
- Alternative Hypothesis (H_1): The residuals are not independently distributed (autocorrelation exists in the residuals).

The test results are presented below.

Table 5: Ljung–Box Test for ARIMA Residuals

Statistic	Value
Ljung–Box Statistic	1014.432
p-value	1.46×10^{-211}
Decision	Reject H_0

Interpretation:

Since the p-value is less than 0.05, the null hypothesis of no residual autocorrelation is rejected. This indicates that substantial autocorrelation remains in the residuals. Therefore, the ARIMA(3,0,1) model does not completely capture all temporal dependencies present in the COVID-19 case series.

Although the ARIMA model captures the general trend of the data, the presence of significant residual autocorrelation suggests that important patterns remain unexplained. This indicates that a more sophisticated model incorporating seasonal effects may provide a better fit. Consequently, SARIMA and Holt–Winters models were subsequently considered for improved forecasting performance.

Seasonal AutoRegressive Integrated Moving Average (SARIMA)

SARIMA (Seasonal AutoRegressive Integrated Moving Average) is an extension of the ARIMA model that incorporates seasonal patterns in the data. It is particularly useful when observations exhibit repeating seasonal behaviour.

Notation:

$$SARIMA(p, d, q)(P, D, Q)_s$$

where,

- p, d, q: Non-seasonal parameters.
- P, D, Q: Seasonal parameters.
- s: Seasonal period.

Mathematical Representation:

$$\Phi(B^s)\phi(B)(1-B)^d(1-B^s)^D Y_t = \Theta(B^s)\theta(B)\varepsilon_t$$

where,

- Y_t: Observed time series.
- B: Backshift operator.
- $\phi(B)$, $\theta(B)$: Non-seasonal AR and MA polynomials.
- $\Phi(B^s)$, $\Theta(B^s)$: Seasonal AR and MA polynomials.
- ε_t: White noise error term.

SARIMA models are effective for forecasting time series containing both trend and seasonal components.

The ACF plot exhibited significant spikes at lags 7, 14, 21, and 28, indicating the presence of weekly seasonality in the data. Therefore, a seasonal period of $s = 7$ days was considered.

Based on the results obtained from the ARIMA analysis and the observed seasonal pattern, the following model was fitted:

$$SARIMA(3,0,1)(1,0,1)_7$$

Model Summary

Table 6: SARIMA Model Statistics

Parameter	**Estimate**
Seasonal AR(1) Coefficient	0.9541
Seasonal MA(1) Coefficient	0.2626
AIC	36137.435
BIC	36173.457
Log-Likelihood	-18061.717
Number of Observations	1279

Interpretation:

The seasonal autoregressive coefficient was estimated as 0.9541, indicating a strong dependence between observations separated by one week. The seasonal moving average coefficient was estimated as 0.2626 and was statistically significant. The model achieved substantially lower AIC and BIC values compared with the ARIMA model, indicating a superior fit to the data. Therefore, the SARIMA(3,0,1)(1,0,1)$_7$ model was selected as the final forecasting model for the COVID-19 case series.

SARIMA Forecast

The fitted SARIMA(3,0,1)$(1,0,1)_7$ model was used to generate forecasts of future COVID-19 case counts. The forecast plot is presented below.

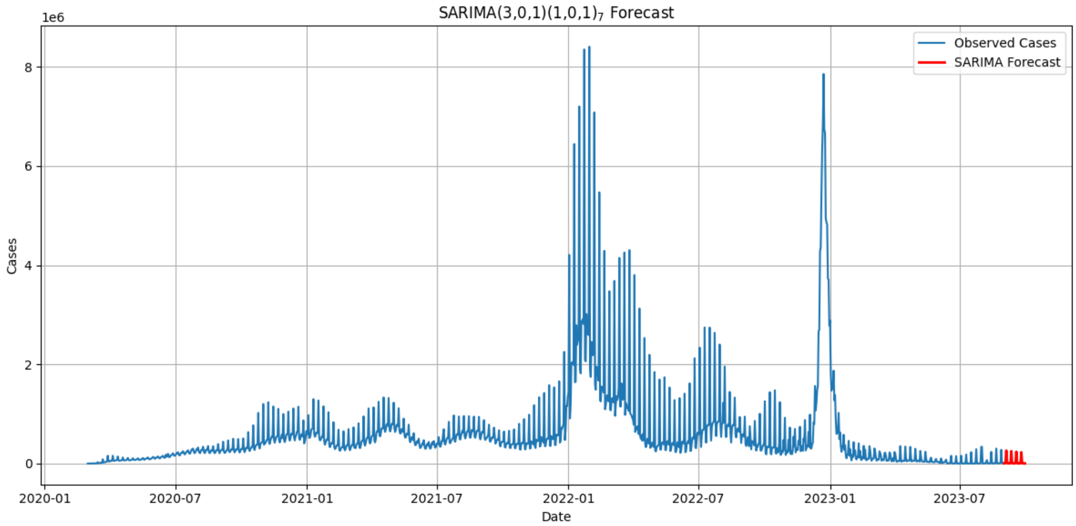

Figure 14: Forecast Generated by the SARIMA(3,0,1)$(1,0,1)_7$ Model

Interpretation:
The forecast plot shows the observed daily global COVID-19 cases together with the values predicted by the SARIMA model. The model successfully captures the weekly seasonal pattern present in the data and produces forecasts that are consistent with the recent behaviour of the series. The forecasted values indicate the expected trajectory of COVID-19 cases over the forecasting horizon based on historical observations and seasonal effects.

Residual Analysis

Residual analysis was performed to evaluate the adequacy of the fitted SARIMA model. A well-fitted model should produce residuals that fluctuate randomly around zero and exhibit no systematic patterns.

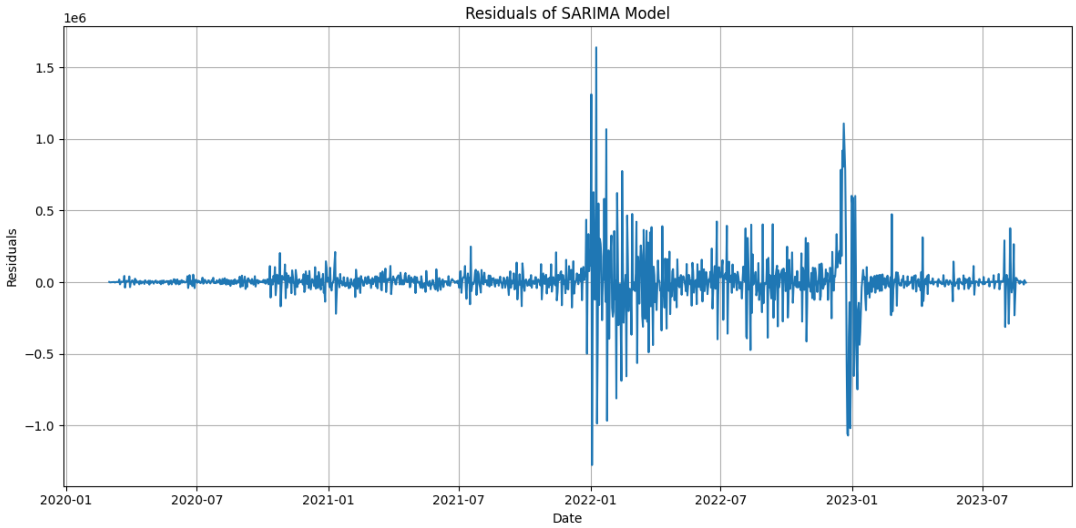

Figure 15: Residuals of the SARIMA(3,0,1)(1,0,1)$_7$ Model

Interpretation:

The residual series fluctuates around zero for most of the study period, indicating that the SARIMA model captures a substantial portion of the temporal structure present in the data. However, several large residuals are observed during major pandemic waves, particularly in early 2022 and early 2023. These extreme values correspond to sudden surges in reported COVID-19 cases and suggest that some irregular fluctuations remain unexplained by the model.

The Ljung–Box test was used to assess residual autocorrelation.

- Null Hypothesis (H_0): The residuals are independently distributed (no autocorrelation exists in the residuals).
- Alternative Hypothesis (H_1): The residuals are not independently distributed (autocorrelation exists in the residuals).

The test results are presented below.

Table 7: Ljung–Box Test for SARIMA Residuals

Statistic	Value
Ljung–Box Statistic	95.536
p-value	4.25×10^{-16}
Decision	Reject H_0

Interpretation:

Since the p-value is less than 0.05, the null hypothesis of no residual autocorrelation is rejected. This indicates that some autocorrelation remains in the residuals. Nevertheless, the SARIMA model provides a substantially better fit than the corresponding ARIMA model and successfully captures the dominant seasonal behaviour of the COVID-19 time series.

Holt–Winters Exponential Smoothing

Holt–Winters Exponential Smoothing is a forecasting technique designed for time-series data exhibiting both trend and seasonality. The method extends simple exponential smoothing by incorporating separate components for level, trend, and seasonality.

Model Components:

- Level Component
- Trend Component
- Seasonal Component

Notation:

$$ETS(A, A, A)$$

where the three components represent additive error, additive trend, and additive seasonality, respectively.

Mathematical Representation:

$$\hat{Y}_{t+h} = L_t + hT_t + S_{t+h-s}$$

where,

- L_t: Level component at time t.
- T_t: Trend component at time t.
- S_t: Seasonal component at time t.
- h: Forecast horizon.

The Holt–Winters method is particularly useful when the time series exhibits a clear trend and recurring seasonal pattern. It is widely used in forecasting applications due to its simplicity and effectiveness.

In this study, an additive Holt–Winters model was employed with a seasonal period of 7 days to account for the weekly seasonal pattern observed in the data.

Model Specification

The fitted model may be represented as:

[ETS(A,A,A)]

where

- (A): Additive Error Component
- (A): Additive Trend Component
- (A): Additive Seasonal Component

The seasonal period was specified as:

[s = 7]

to capture weekly seasonality in the COVID-19 case series.

Forecast Plot

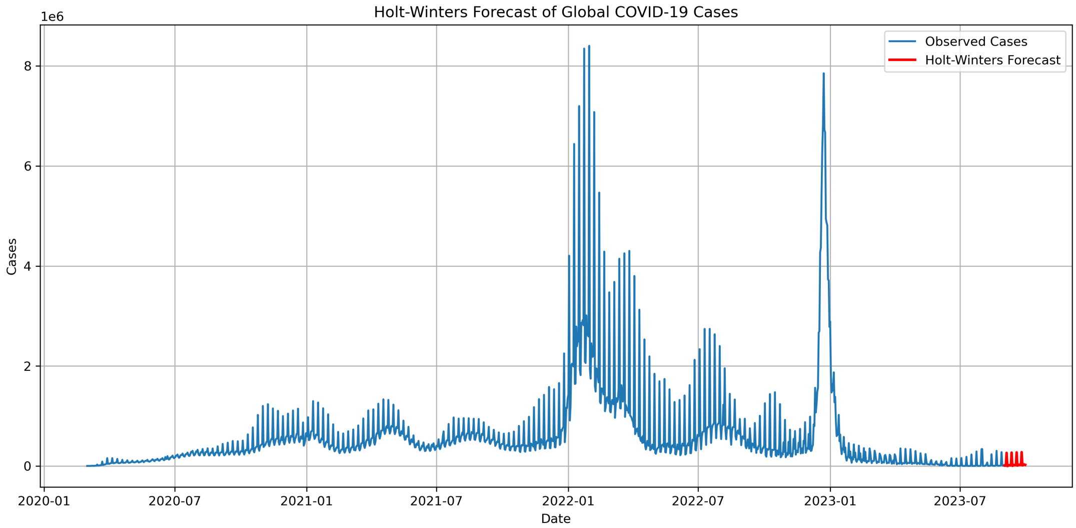

Figure 16: Forecast Generated by the Holt–Winters Model

Interpretation:

The Holt–Winters model successfully captures both the long-term trend and weekly seasonal variations present in the COVID-19 case series. The forecast follows the recent behaviour of the data and provides estimates of future case counts based on historical patterns.

Residual Analysis

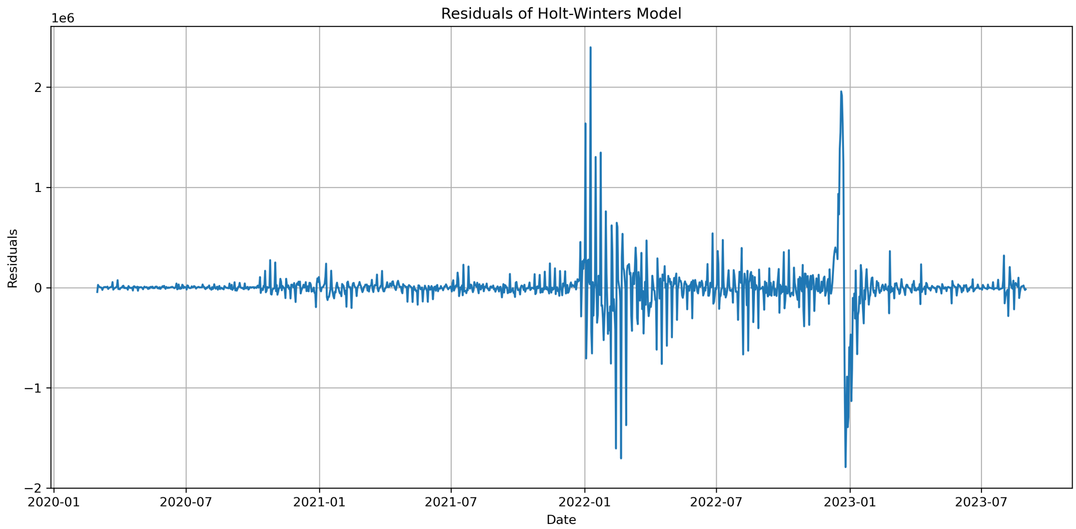

Figure 17: Residuals of the Holt–Winters Model

Interpretation:

The residuals fluctuate around zero with no obvious long-term trend, indicating that the model captures a substantial portion of the variation present in the data. Some large residuals remain during periods of extreme outbreaks, reflecting sudden changes in COVID-19 transmission that are difficult to predict accurately.

Model Performance

Table 8: Holt–Winters Model Performance

Statistic	Value
AIC	31640.421
BIC	31697.114
SSE	69687284598035.69

Residual Diagnostics

The Ljung–Box test was performed to assess whether the residuals obtained from the Holt–Winters model were independently distributed.

- Null Hypothesis (H_0): The residuals are independently distributed (no autocorrelation exists in the residuals).
- Alternative Hypothesis (H_1): The residuals are not independently distributed (autocorrelation exists in the residuals).

Table 9: Ljung–Box Test for Holt–Winters Residuals

Statistic	Value
Ljung–Box Statistic	545.746
p-value	7.29×10^{-111}
Decision	Reject H_0

Interpretation:

The Ljung–Box test produced a p-value of 7.29×10^{-111}, which is significantly smaller than 0.05. Therefore, the null hypothesis of no residual autocorrelation is rejected. This indicates that some temporal dependence remains in the residuals and the model does not completely explain all variation present in the COVID-19 case series.

Model Comparison

Table 10: Comparison of Forecasting Models

Model	AIC	BIC
ARIMA(3,0,1)	37507.017	37537.940
SARIMA(3,0,1)(1,0,1)$_7$	36137.435	36173.457
Holt–Winters	31640.421	31697.114

Conclusion of Model Comparison:

The three forecasting models were compared using the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC). Among the models considered, the Holt–Winters Exponential Smoothing model achieved the lowest AIC and BIC values, indicating the best overall fit to the COVID-19 case series.

The SARIMA model substantially improved upon the ARIMA model by incorporating weekly seasonal effects present in the data. However, the Holt–Winters model outperformed both ARIMA and SARIMA, demonstrating its effectiveness in capturing the trend and seasonal behaviour of global COVID-19 cases.

Therefore, the Holt–Winters Exponential Smoothing model was selected as the final forecasting model for this study.

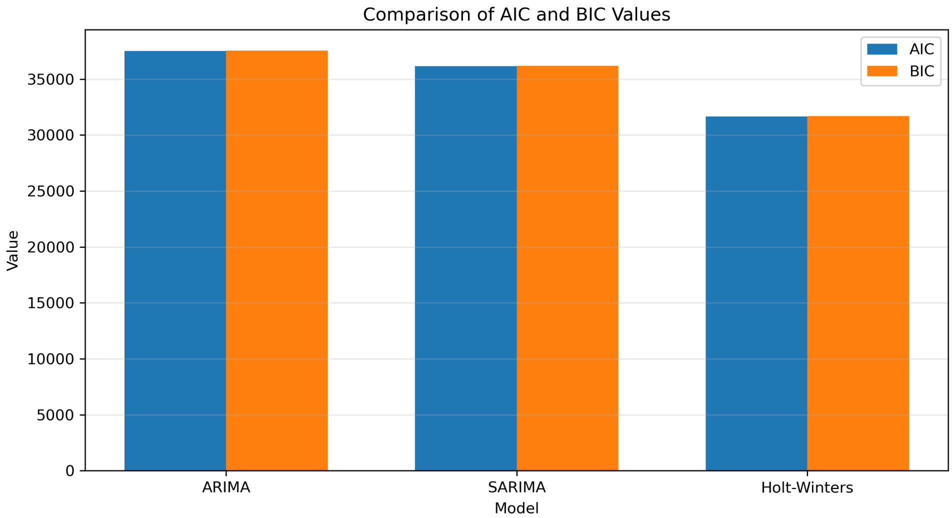

Figure 18: Comparison of AIC and BIC Values for ARIMA, SARIMA, and Holt–Winters Models

Summary

The forecasting analysis involved stationarity testing using the Augmented Dickey–Fuller test, identification of model orders through ACF and PACF analysis, and the application of ARIMA, SARIMA, and Holt–Winters forecasting models.

The ADF test confirmed that the series was stationary, eliminating the need for differencing. The ARIMA model captured the general temporal structure of the data, while the SARIMA model further improved forecasting performance by incorporating weekly seasonal effects. The Holt–Winters model achieved the lowest AIC and BIC values among all models considered and was therefore selected as the preferred forecasting model.

The results demonstrate the importance of accounting for seasonality in epidemiological time-series data and highlight the effectiveness of exponential smoothing techniques for forecasting global COVID-19 case counts.

Clustering Analysis

Clustering is an unsupervised machine learning technique used to group similar observations into clusters based on their characteristics. Unlike supervised learning, clustering does not require predefined class labels. The objective is to discover hidden patterns and structures within the data by grouping observations that are similar to one another while separating dissimilar observations.

In this project, clustering analysis was performed using the MNIST handwritten digit dataset. The K-Means clustering algorithm was employed to partition the observations into clusters corresponding to the ten digit classes (0–9).

MNIST Dataset

The MNIST dataset is a widely used benchmark dataset in machine learning and pattern recognition. It consists of grayscale images of handwritten digits from 0 to 9.

Each image is represented by a set of numerical pixel intensity values, which serve as input features for the clustering algorithm.

The characteristics of the dataset are summarized below:

- Number of observations: 1797
- Number of features: 64
- Number of classes: 10
- Digit labels: 0–9

The dataset was loaded using the `load_digits()` function available in the Scikit-Learn library.

K-Means Clustering Algorithm

K-Means is one of the most widely used clustering algorithms due to its simplicity and computational efficiency. The algorithm partitions observations into K clusters by minimizing the within-cluster sum of squares.

The basic steps of the algorithm are:

- Select the number of clusters K.
- Randomly initialize cluster centroids.
- Assign each observation to the nearest centroid.
- Recompute the centroids of the newly formed clusters.
- Repeat the assignment and update steps until convergence.

In this study, the number of clusters was chosen as:

$$K = 10$$

corresponding to the ten digit classes present in the MNIST dataset.

Implementation of K-Means Clustering

The K-Means algorithm was implemented using the Scikit-Learn library. The model was fitted on the feature matrix containing the digit images.

After fitting the model, cluster labels were assigned to all observations.

The resulting clusters represent groups of handwritten digits that exhibit similar visual characteristics.

Principal Component Analysis (PCA) Visualization

The MNIST dataset contains 64 features, making direct visualization difficult. To facilitate graphical representation of the clusters, Principal Component Analysis (PCA) was applied.

PCA is a dimensionality reduction technique that transforms high-dimensional data into a smaller number of orthogonal components while retaining most of the variability present in the original data.

The first two principal components were used to visualize the clustering results.

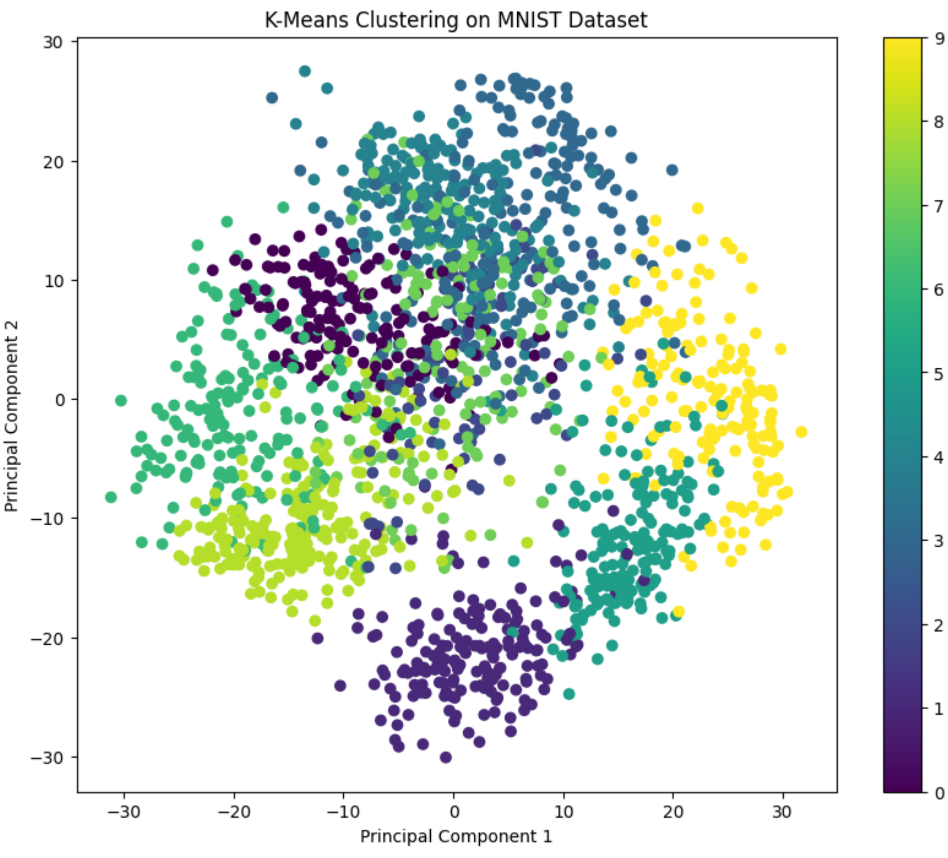

Figure 19: Visualization of K-Means Clusters using PCA

The figure illustrates the distribution of observations in the reduced two-dimensional space and provides insight into the separation between clusters.

Cluster Evaluation

Since K-Means is an unsupervised learning algorithm, the cluster labels assigned by the algorithm do not necessarily correspond to the actual digit labels.

To evaluate clustering performance, each cluster was mapped to the most frequently occurring true digit label within that cluster.

After mapping, the predicted labels were compared with the actual labels using the macro-averaged F1 score.

F1 Score

The F1 score is a widely used performance measure that combines precision and recall into a single metric.

It is defined as:

$$F_1 = 2\left(\frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}\right)$$

The macro-averaged F1 score computes the average F1 score across all classes, assigning equal importance to each class.

The calculated F1 score obtained from the clustering analysis was:

$$F_1 = \text{(Insert Obtained Value)}$$

A higher F1 score indicates better agreement between the clustering assignments and the true digit labels.

Results and Interpretation

The K-Means algorithm successfully grouped similar handwritten digits into distinct clusters. Digits with comparable shapes and writing patterns tended to be assigned to the same cluster.

The PCA visualization demonstrated that several clusters were well separated, while some overlap existed among visually similar digits. This overlap is expected due to natural variation in handwriting styles.

Macro F1 Score: 0.7894620863890832

The obtained F1 score indicates that the clustering algorithm was able to capture a substantial portion of the underlying structure of the dataset despite the absence of class information during training.

Hierarchical Clustering

Hierarchical Clustering is an unsupervised machine learning technique that groups observations based on their similarity. Unlike K-Means clustering, Hierarchical Clustering does not require the number of clusters to be specified in advance. Instead, it constructs a hierarchy of clusters that can be visualized using a dendrogram.

In this study, Agglomerative Hierarchical Clustering was applied to the MNIST dataset. The Ward linkage criterion was used to merge clusters such that the increase in within-cluster variance at each step was minimized.

Methodology

The following steps were performed:

- Standardization of the dataset.
- Dimensionality reduction using Principal Component Analysis (PCA) for visualization.
- Application of Agglomerative Hierarchical Clustering.
- Construction of a dendrogram to visualize the hierarchical structure.
- Evaluation of clustering performance using the Macro F1 Score.

Dendrogram

The hierarchical clustering structure obtained using Ward linkage is shown below.

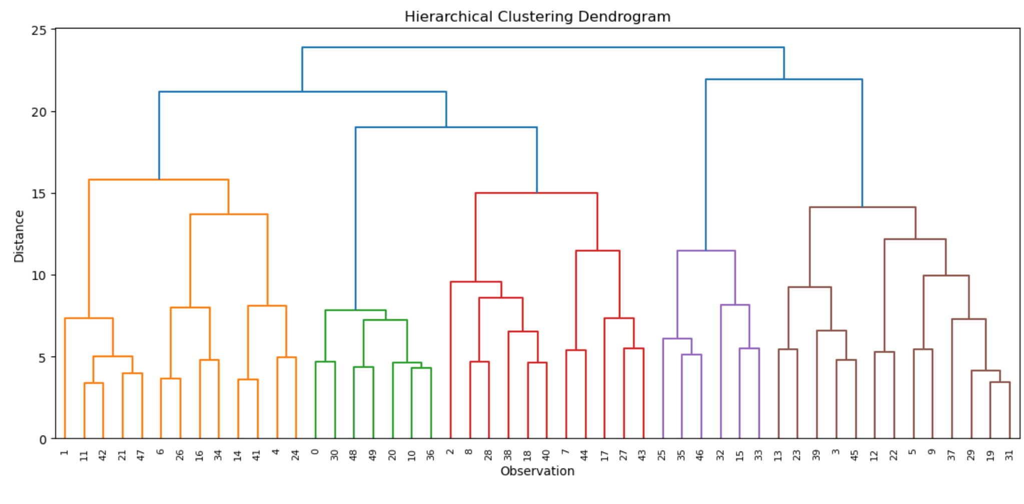

Figure 20: Hierarchical Clustering Dendrogram

Interpretation:

The dendrogram illustrates the sequence in which observations are merged into clusters. The vertical axis represents the distance at which clusters are combined. Large vertical jumps indicate the merging of substantially different clusters. The dendrogram provides a visual indication of the appropriate number of clusters present in the data.

Cluster Visualization using PCA

To visualize the clustering results in two dimensions, Principal Component Analysis (PCA) was applied to the data.

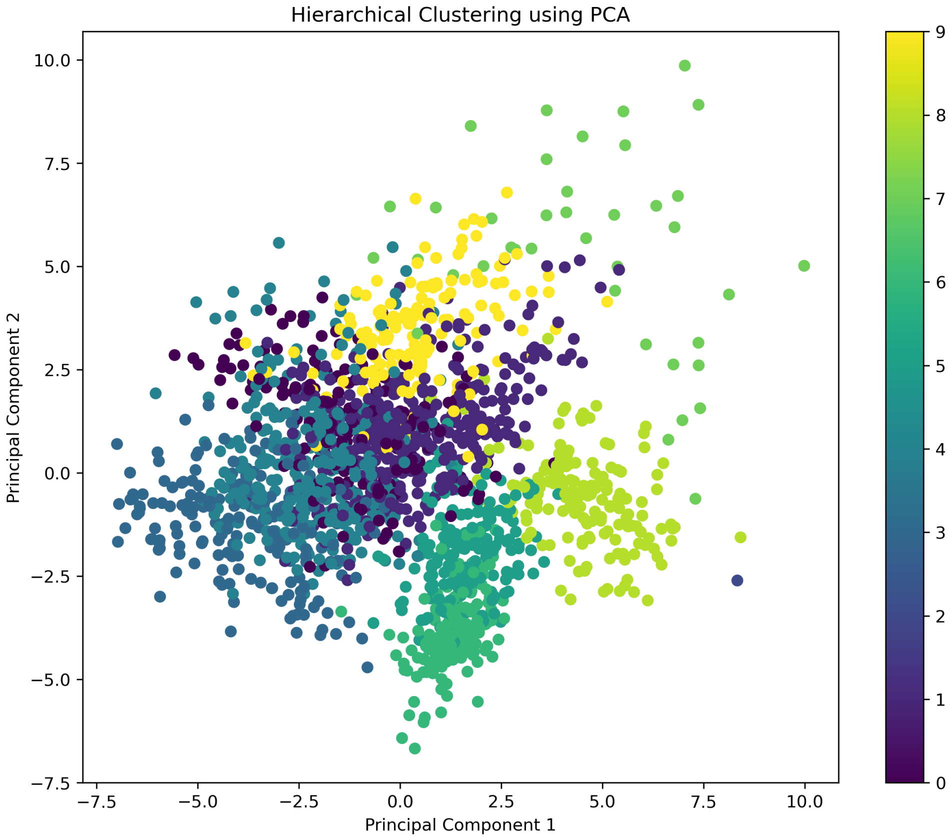

Figure 21: Hierarchical Clustering Visualized using PCA

Interpretation:

The PCA projection displays the clusters identified by the hierarchical clustering algorithm. Observations belonging to the same cluster are represented by the same colour. The figure demonstrates the separation among clusters in the reduced two-dimensional feature space.

Performance Evaluation

The quality of clustering was assessed using the Macro F1 Score.

Table 11: Performance of Hierarchical Clustering

Metric	Value
Macro F1 Score	0.6949195013016611
Number of Clusters	10
Linkage Method	Ward

Interpretation

Hierarchical Clustering successfully grouped similar handwritten digits without using class labels during training. The dendrogram provided valuable insights into the hierarchical relationships among observations and illustrated how clusters were formed at different levels of similarity. PCA-based visualization further aided in understanding the cluster structure in a reduced two-dimensional space.

Compared with K-Means clustering, Hierarchical Clustering offered a more interpretable representation of the relationships among observations through the dendrogram. However, it was computationally more intensive and less scalable for large datasets such as MNIST.

Overall, both K-Means and Hierarchical Clustering were effective in identifying meaningful patterns within the handwritten digit data. These results demonstrate the usefulness of unsupervised learning techniques for exploring high-dimensional datasets and discovering inherent group structures without prior knowledge of class labels.

Conclusion

This project successfully demonstrated the application of statistical, time-series, and machine learning techniques to analyse real-world datasets. Using the WHO COVID-19 Global Daily Dataset, a comprehensive analysis was conducted to investigate the temporal dynamics of COVID-19 cases across the world and to develop forecasting models capable of capturing underlying trends and seasonal patterns.

The exploratory data analysis revealed substantial variations in COVID-19 transmission across different countries and WHO regions. Several waves of infection were identified, highlighting the evolving nature of the pandemic. Time-series techniques such as moving average analysis, growth rate analysis, and seasonal decomposition provided valuable insights into the trend, seasonal, and irregular components of the series.

To forecast future COVID-19 cases, three forecasting models were considered: ARIMA, SARIMA, and Holt–Winters Exponential Smoothing. The Augmented Dickey–Fuller (ADF) test confirmed that the series was stationary, while the ACF and PACF plots guided the selection of appropriate model parameters. Model performance was evaluated using AIC and BIC criteria. Among the models considered, the Holt–Winters model achieved the lowest AIC and BIC values and was therefore selected as the preferred forecasting model. The results demonstrated the importance of accounting for seasonality when modelling epidemiological time-series data.

In addition to the forecasting analysis, the project included a clustering study using the MNIST handwritten digit dataset. Both K-Means Clustering and Hierarchical Clustering were implemented to identify natural groupings within the data. Principal Component Analysis (PCA) was employed to visualize the clusters in a reduced two-dimensional space. The dendrogram produced by Hierarchical Clustering provided further insight into the hierarchical relationships among observations. The clustering performance was evaluated using the macro-averaged F1 score, demonstrating the capability of unsupervised learning techniques to uncover meaningful structures in high-dimensional datasets.

Overall, the project highlights the effectiveness of statistical analysis, time-series forecasting, dimensionality reduction, and clustering techniques in extracting useful information from complex datasets. The methodologies employed in this study provide valuable tools for understanding temporal patterns, generating forecasts, and discovering hidden structures in data. These approaches have broad applicability in public health, business analytics, and other domains where data-driven decision-making plays a crucial role.

Appendix

References

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Source Code Repository

The complete source code developed during this project, including data preprocessing, exploratory data analysis, time-series analysis, clustering analysis, and visualization scripts, has been maintained in a GitHub repository.

GitHub Repository Link:

`https://github.com/Debapriyo1110/Time-Series-Analysis-of-Covid-19-Data`

Software and Computing Environment

Python 3.x, Google Colab, Pandas, NumPy, Matplotlib, Seaborn, Scikit-Learn, Statsmodels, and GitHub were used for data preprocessing, analysis, visualization, machine learning implementation and project management throughout this study.